



REVIEW

Is a Functional Cure Possible in Autoimmune Diseases? Evidence from Trigger Eradication, Transplantation, and Cellular Therapies

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Received: October 18, 2025 / Accepted: December 3, 2025 / Published online: December 23, 2025
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ABSTRACT

Introduction: Traditionally considered incurable, autoimmune diseases (AIDs) may—in specific circumstances—achieve sustained remission or even a “functional cure,” defined as durable clinical and laboratory remission without immunosuppression. This review evaluates evidence across five therapeutic axes: infectious trigger eradication, immune reset via autologous hematopoietic stem cell transplantation (HSCT), cellular therapies (CAR-T, extracorporeal photopheresis), environmental/nutritional strategies, and paraneoplastic syndromes.

Methods: Systematic review according to PRISMA guidelines in PubMed/MEDLINE, Embase, Scopus, Web of Science, and Cochrane (up to September 2025). Eligible studies included trials, meta-analyses, cohorts, case series, and

reports describing sustained or drug-free remission. Definitions applied were clinical remission, complete remission, sustained remission ≥ 12 months, drug-free remission, and functional cure (complete, off-therapy remission with stable biomarkers and no new organ damage).

Results: Strong evidence supports *Helicobacter pylori* eradication in immune thrombocytopenic purpura, with signals in dermatoses and urticaria. In systemic sclerosis, HSCT outperformed cyclophosphamide in randomized trials, improving survival and reducing prolonged immunosuppression; lupus series reported extended drug-free remissions. Anti-CD19 CAR-T therapies induced deep remission in B-cell-mediated AIDs, normalizing autoantibodies over 12–24 months. Photopheresis showed safety but heterogeneous efficacy. Environmental interventions (vitamin D, plant-based diet, microbiota modulation) suggested benefit, though with limited evidence for cure. In paraneoplastic syndromes, tumor control often coincided with autoimmune remission.

Conclusions: Functional cure in AIDs appears achievable in selected cases through trigger removal, immune reset, or profound immune depletion. Advancing this paradigm requires standardized definitions, predictive biomarkers, and long-term controlled trials to integrate these strategies into routine care.

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Keywords: Autoimmune diseases; Functional cure; Sustained remission; *Helicobacter pylori*; Autologous hematopoietic stem cell transplantation; CAR-T; Extracorporeal photopheresis; Paraneoplastic syndromes; Vitamin D; Plant-based diet

Key summary points

Functional cure in autoimmune diseases is possible in selected contexts through immune tolerance restoration.

Helicobacter pylori eradication in ITP demonstrates sustained drug-free remission via trigger removal.

Hematopoietic stem cell transplant (HSCT) and CAR-T therapies can reset the immune system and induce long-term remission.

Nutritional and environmental interventions, such as vitamin D and plant-based diets, support immune modulation.

Paraneoplastic syndromes show remission after tumor treatment, reinforcing antigen elimination as a therapeutic principle.

INTRODUCTION

Autoimmune diseases (AIDs) are a diverse group of conditions characterized by the breakdown of immune tolerance against self-antigens with ensuing chronic inflammation and progressive tissue injury [1, 2]. Traditionally, these diseases are thought to be incurable with treatment designed primarily at controlling the inflammatory activity aside from preventing irreversible damage and securing quality of life [3]. The natural course of many diseases, such as systemic lupus erythematosus (SLE), rheumatoid arthritis (RA), systemic sclerosis (SSc), and immune thrombocytopenic purpura (ITP), has been well described to be chronic and relapsing, fluctuating between remission and flare-up, rather than resolving completely [4, 5].

There is growing evidence in recent decades that a ‘functional cure’ may be possible under specific conditions. This idea was first introduced in infectious diseases like HIV and is defined as a state of antiretroviral therapy (ART)-free HIV-infected patients who are asymptomatic and clinically stable, with normalization or stabilization of biomarkers for immunological activity [6]. When translated to autoimmunity would be the long-term clinical remission, without any immunosuppressive drugs and normalization of serological features (auto-antibodies and complement) [7]. In this review, we use the term “functional cure” in this operational sense, acknowledging that it does not imply complete eradication of genetic or immunologic susceptibility to disease relapse.

The plausibility of this phenomenon is supported by a number of clinical models. Among the most established models is clearance of infectious triggers like *Helicobacter pylori* with eradication leading to a complete or durable response in patients with ITP [8, 9]. A similar model applies to paraneoplastic syndromes, where curative treatment of the underlying cancer often causes remission of associated autoimmune findings (e.g., TIF1 γ -related dermatomyositis [10, 11]).

Another area of interest is immune reconstitution after intense therapeutic regimens, such as autologous-hematopoietic stem cell transplant (HSCT), and most recently with advanced cellular therapies of chimeric antigen receptor T cells (CAR-T) [12, 13]. Randomized controlled trials in SSc have shown that HSCT may lead to durable remission and reduce the requirement for long-term immunosuppressive medication [14, 15]. Likewise, recent series on the use of CAR-T cells in both SLE and other such B-cell-mediated autoimmune diseases have shown remissions extending over 1 year off intervention [16, 17].

Besides the immunological approach, evidence for the influence of environmental and nutritional factors to modify autoimmunity is increasing. For instance, vitamin D functions as an immune response modulating agent that has been linked to down-regulation of the activity in an autoimmune disease [18]. Similarly, unprocessed vegan diets can contribute to persistent

remission in SLE and Sjögren's syndrome according to research [19].

Extracorporeal photopheresis (ECP) likewise appears as a promising immunomodulatory approach. While clinical trials in systemic sclerosis have failed to show clear-cut superiority over standard treatment, ECP is still a valuable physiologic model of immunologic tolerance induction attempts [20].

The purpose of this review is to describe at length the evidence on sustained remission and functional cure in ADs, breaking it down into five major axes: (1) eradication of infectious triggers, (2) immune reset through chemotherapy or autologous hematopoietic stem cell transplantation (aHSCT), (3) state-of-the-art cellular therapies; (4) Advanced environmental and nutritional approaches; and (5) remissions that happen within a cancer treatment setting as seen in paraneoplastic conditions. We deliberately focused on these five therapeutic axes—*infectious trigger eradication, immune reset via HSCT or cytotoxic regimens, advanced cellular therapies, environmental and nutritional strategies, and paraneoplastic contexts*—because each provides mechanistic insight into reversal of autoimmunity and sustained drug-free remission. Conventional synthetic and biological disease-modifying antirheumatic drugs (DMARDs) are essential for disease control and can lead to DMARD-free sustained remission in a subset of patients with rheumatoid arthritis and other diseases, but they have been extensively reviewed elsewhere and generally act as long-term suppressive rather than truly resetting interventions.

METHODS

The present systematic review followed the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [21]. The search strategy was conducted in PubMed/Medline, Embase, Scopus, Web of Science, and the Cochrane Library from their inception until September 2025 with no language restrictions applied. MeSH terms and key words were modified for each database but included five

major axes: (1) eradication of infectious triggers including *Helicobacter pylori*, viruses or other bacteria; (2) immunological reset achieved through chemotherapy schemes or autologous HSCT; (3) advanced cell therapies comprising CAR-T, extracorporeal photopheresis so on; (4) environmental and nutritional interventions such as vitamin D supplementation, eating habits modification, microbiota manipulation; and paraneoplastic syndromes in which cancer treatment can lead to disappearance of the cause-specific autoimmune artifact.

Inclusion criteria: Eligible studies were RCTs, meta-analysis, systematic reviews, observational (cohort and case-control), case series, and individual case reports reporting sustained remission or functional cure in autoimmune diseases. Only studies with a clear definition of sustained clinical remission, typically ≥ 6 months, were included and preferably trial discontinuation of immunosuppressive therapy (IST), with or without laboratory evidence for remission. Papers reporting partial improvement of disease activity without meeting remission criteria, follow-up less than 6 months, or those that did not report objective clinical and laboratory data were excluded.

One pivotal aspect of this review was to standardize the conceptual definitions being used: “Clinical remission” was determined as the absence of clinical signs and symptoms of disease activity by internationally established criteria for each disorder, including Disease Activity Score Calculator for Rheumatoid Arthritis (DAS28) < 2.6 for rheumatoid arthritis [22]. The Systemic Lupus Erythematosus Disease Activity Index 2000 (SLEDAI-2 K) = 0 for systemic lupus erythematosus [23], and stability of modified Rodnan skin score (mRSS) in systemic sclerosis [24]. “Complete remission” was defined as resolution of clinical symptoms and normality of all laboratory tests, including either negativity or stable values for autoantibodies, complement, and inflammatory markers. “Sustained remission” was defined as the persistence of these criteria for a minimum of 12 months. “Drug-free remission” referred to the maintenance of sustained remission off immunosuppressive agents and corticosteroids. Finally, “functional cure” was defined as the

same condition as complete clinical remission for at least 12 months without immunosuppressive therapy, but also normal immunological biomarkers or stabilization of them and no new appearance of progressive damage during follow-up. These definitions were adopted from the respective international consensus papers: DORIS, for lupus [23], EULAR/ACR recommendations, for RA [22], and EUSTAR guidelines, for SSc [24] and applied transversally to all AD to make comparison across them viable. We recognize that not all primary studies used exactly these criteria; when different remission definitions were reported, we mapped their endpoints onto this framework (clinical, complete, sustained, drug-free remission or functional cure) as closely as possible and clarified these details in the corresponding tables.

Three steps were used in the selection of articles. Firstly, we screened titles and abstracts to remove studies that were obviously irrelevant. Full paper reading was then conducted for potentially relevant articles. Any disagreements between reviewers were settled by consensus. For each identified study, we extracted data on: type of study design (randomized controlled clinical trial [RCT], non-randomized, cohort, systematic review or case report), number of patients, characteristics of the population studied (underlying disease/condition and treatment received if any e.g., *H. pylori* eradication, HSCT and CAR-T; diet vs. cancer therapy), criteria used to define remission after fecal microbiota transplantation (FMT) intervention(s), reported response rates to FMTs administered long-term post-FMT follow-up duration adverse events including associated mortality.

Results were synthesized in a narrative and thematic structure along the five axes we had established. The CDS to foster an interpretation by which data collected in separate fields (infectious, immunological, cellular, environmental, and oncologic) were once again seen under one concept: the reversibility of autoimmunity and the restoration of self-tolerance.

This article is based on previously conducted studies and does not contain any new studies with human participants or animals performed by any of the authors.

Additional methodological details and extended content are available in the Supplementary Material.

RESULTS

Eradication of Infectious Triggers

The best-described model of long-standing remission in autoimmune diseases following eradication therapy is infection with *Helicobacter pylori* (Table 1). It is in ITP that the closest association has been demonstrated and where the microorganism appears to act as a cofactor in both its induction and perpetuation of autoreactive immune response against platelets. Meta-analyses have reported that approximately half of the patients demonstrate a sustained increment in platelet count after eradication treatment, and some may even achieve complete normalization without any kind of further therapy [8, 9, 25, 26]. The effect is also highly significant in Asian and European populations, indicating that genetic and immunogenetic factors (Fcγ-receptor polymorphisms) could influence therapeutic effects in addition to Fc receptor signaling molecules, such as the Fcγ receptors, to some variants of bacterial antigens in recognition [27].

The principal mechanism suggested is molecular mimicry, i.e., *H. pylori* epitopes have structural similarity with platelet glycoproteins (GPIIb/IIIa) resulting in an autoreactive response. The removal of the antigenic stimulus by the bacterial eradication permits for regulatory mechanisms like regulatory T cells (Tregs) and anti-inflammatory cytokines (e.g., IL-10) to re-establish peripheral tolerance [28, 29].

Comparable phenomena have been described in autoimmune skin disorders. Improvement of or even disappearance of lesions in palmoplantar psoriasis, plaque psoriasis, and oral lichen planus after *H. pylori* eradication suggests that the pathogen serves as a sustaining factor for cutaneous inflammation [30, 31]. In Behçet's disease, case reports described the decreased frequency of oral and genital ulcers and ocular involvement following antibacterial treatment

Table 1 Eradication of infectious triggers associated with sustained remission in autoimmune diseases

Author/Year [Ref.]	Infectious agent/ Intervention	Autoimmune disease	Study type	No. of patients	Follow-up	Remission/clinical outcome criteria	Main results	Comments/ adverse events
Stasi et al. 2009 [8]	<i>Helicobacter pylori</i> eradica- tion (triple therapy)	Immune throm- bocytopenic purpura (ITP)	Meta-analysis (17 studies)	788	6–24 months	Sustained platelet count increase ($> 100,000/\text{mm}^3$)	51% achieved complete or partial response; higher response rates in Asian and European populations	Mild gastrointes- tinal events; no severe adverse effects
Franchini et al. 2007 [9]	<i>H. pylori</i> eradica- tion therapy	ITP	Systematic review	788	Variable	Complete response rate	Mean CR rate 42%; strongest association with CagA + strains	Evidence suggests molecular mimicry with platelet glyco- proteins
Emilia et al. 2007 [25]	<i>H. pylori</i> eradica- tion (clarithro- mycin-based)	ITP	Prospective cohort	40	12 months	Sustained platelet nor- malization without therapy	63% achieved complete remission; maintained up to 1 year	No relapse dur- ing follow-up
Michel et al. 2011 [26]	<i>H. pylori</i> eradica- tion	ITP	Multicenter retrospective	280	24 months	Platelet response $> 30 \times 10^9/l$	43% responders; higher efficacy with early eradication	Confirms causal relationship hypothesis
Veneri et al. 2005 [27]	<i>H. pylori</i> eradica- tion	ITP	Case series	28	12 months	Platelet normalization	50% achieved complete remis- sion	Suggested modu- lation of Th1/ Th2 balance

Table 1 continued

Author/Year [Ref.]	Infectious agent/ Intervention	Autoimmune disease	Study type	No. of patients	Follow-up	Remission/clinical outcome criteria	Main results	Comments/ adverse events
Campuzano- Maya 2014 [30]	<i>H. pylori</i> eradica- tion	Psoriasis (vul- garis, palmo- plantar)	Case-control	30	6–12 months	PASI improve- ment $\geq 75\%$	53% achieved complete or near-complete skin lesion remission	Supports bacte- rial antigen persistence hypothesis
Ergun et al. 2010 [31]	<i>H. pylori</i> eradica- tion	Oral lichen planus	Case-control	40	6 months	Clinical remission of oral lesions	60% achieved partial or com- plete remission	Suggests <i>H. pylori</i> 's role as chronic anti- genic stimulus
Akdeniz et al. 2015 [32]	<i>H. pylori</i> eradica- tion	Behçet's disease	Case series	22	12 months	Decrease in oral/geni- tal ulcer frequency	68% achieved significant improvement; ocular symp- toms reduced	No serious AEs; effect sustained at 12 months
Magen et al. 2016 [33]	<i>H. pylori</i> eradica- tion	Chronic sponta- neous urticaria	Case series	18	6 months	Resolution of urticaria symptoms	83% complete remission; no recurrence reported	Proposed restora- tion of immune tolerance post- eradication
Bassi et al. 2017 [34]	<i>H. pylori</i> eradica- tion	Autoimmune thyroiditis/ Sjögren's syn- drome	Case reports (2)	2	6–9 months	Reduction of autoan- tibody titers/clinical improvement	Both cases showed clini- cal stabiliza- tion; partial autoantibody normalization	Limited evi- dence; hypoth- esis-generating only

Table 1 continued

Author/Year [Ref.]	Infectious agent/ Intervention	Autoimmune disease	Study type	No. of patients	Follow-up	Remission/clinical outcome criteria	Main results	Comments/ adverse events
Terao et al. 2018 [35]	Antiviral therapy (DAAs) for hepatitis C	Cryoglobulinemic vasculitis	Multicenter cohort	120	24 months	Sustained virologic response + remission of vasculitis	78% achieved complete remission; relapse rare after SVR	Demonstrates link between viral eradication and immune reset
Balfour et al. 2020 [36]	Antiviral therapy for EBV-associated disorders	Secondary autoimmune diseases (various)	Review	–	–	Resolution of autoimmune manifestations post-viral control	Remission correlates with viral suppression or clearance	Highlights infection-driven autoimmunity paradigm

ITP immune thrombocytopenic purpura, *CR* complete response, *PAS/Psoriasis* Area and Severity Index, *SVR* sustained virologic response, *EBV* Epstein–Barr virus, *DAAs* direct-acting antivirals

[32]. These factors lend support to the theory that infectious agents could act as chronic antigenic stimulants, perpetuating continual immune activation.

Less commonly, however, other autoimmune disorders can resolve after bacterial eradication. Chronic spontaneous urticaria (CSU) episodes have resolved in cases following *H. pylori* eradication [33]. There are, however, cases in which autoimmune thyroiditis and Sjögren’s syndrome seem to improve following bacterial eradication; unfortunately, very little evidence supports this data, as there is almost no RCT [34].

These data imply that in a subset of autoimmune diseases, disease activity requires the constant presence of infectious triggers. Therefore, elimination of the provoking agent might lead to restoration of immune tolerance—a “functional cure”. This also applies to other infectious conditions, e.g., hepatitis C and Epstein–Barr virus, where antiviral suppression or clearance has led to the disappearance of secondary autoimmune manifestations [35, 36].

Immune Reset: Hematopoietic Stem Cell Transplantation (HSCT)

Autologous hematopoietic stem cell transplantation (HSCT) is one of the most extensively used immunoablative approaches to achieve long-term remission in severe and/or refractory autoimmune diseases. The key principle is the depletion of autoreactive lymphocytic clonal populations with ablative immunosuppression and reconstitution (rebooting) of a new immune repertoire by the infusion of HSCs, thereby recreating both central and peripheral tolerance [12, 13, 36] (Table 2).

The superiority of HSCT over conventional immunosuppressive treatment was demonstrated in the ASTIS (Autologous Stem Cell Transplantation International Scleroderma) and SCOT (Scleroderma: Cyclophosphamide or Transplantation) trials that were performed in patients with diffuse systemic sclerosis. In the ASTIS trial, event-free survival was 79% in transplanted patients and 51% in controls after 5 years [12]. These results were confirmed by the SCOT study, where 8-year overall survival was

Table 2 Hematopoietic stem cell transplantation (HSCT) as an immune reset strategy in autoimmune diseases

Author/Year [Ref.]	Autoimmune disease	Study type	No. of patients	Follow-up	Conditioning regi- men/Protocol	Remission/ Clinical out- come criteria	Main results	Comments/ adverse events
van Laar et al. 2014 (ASTIS Trial) [12]	Systemic scler- osis (diffuse subtype)	Randomized controlled trial	156	5 years	High-dose cyclophospha- mide + ATG	Event-free sur- vival; mRSS, FVC improve- ment	Event-free survival 79% (HSCT) vs. 51% (CYC); improved skin and lung func- tion	Treatment-related mortality 10%; improved long- term survival
Sullivan et al. 2018 (SCOT Trial) [13]	Systemic scler- osis	Randomized controlled trial	75	8 years	Total-body irradiat- ion + CYC + ATG	Global survival; progression- free survival	Overall survival 79% vs. 51% (CYC group); significant quality-of-life improvement	Early mortality 3%; sustained clinical benefit
Jayne et al. 2004 [14]	Systemic lupus erythematosus (SLE)	Multicenter cohort	50	10 years	Cyclophospha- mide + ATG (non- myeloablative)	Sustained clini- cal and sero- logic remission	60% achieved drug-free remis- sion > 10 years; normalization of complement and anti- dsDNA	Infection-related mortality 6%; durable immune recon- stitution
Alexander et al. 2009 [15]	SLE (refractory)	Prospective observational	26	8 years	CYC + ATG + stem cell rescue	Drug-free remis- sion	54% complete remission, anti-dsDNA negativity, complement normalization	Long-term drug- free remissions observed

Table 2 continued

Author/Year [Ref.]	Autoimmune disease	Study type	No. of patients	Follow-up	Conditioning regi- men/Protocol	Remission/ Clinical out- come criteria	Main results	Comments/ adverse events
Burt et al. 2012 [39]	Multiple scler- osis	Randomized phase II trial	110	5 years	Intermediate- intensity (BEAM) + ATG	No evidence of disease activity (NEDA)	78% achieved NEDA status; superior to natalizumab	Lower relapse and disability rates; accept- able toxicity
Farge et al. 2020 (EULAR Reg- istry) [41]	Autoimmune diseases (vari- ous)	Registry-based cohort	1,100	10 years	CYC + ATG (stand- ardized European protocol)	Survival; drug- free remission	TRM reduced from 11% (1997–2000) to < 2% (2015–2019); long-term remission in the majority	Demonstrates protocol optimization and improved safety
Muraro et al. 2017 [43]	Multiple scler- osis	Phase II clinical trial	120	7 years	BEAM + ATG	NEDA; relapse- free survival	> 70% achieved sustained remis- sion > 5 years	Supports the immune reconstitution hypothesis
Oyama et al. 2019 [42]	Crohn's disease	Phase I/II trial	24	4 years	Cyclophospha- mide + ATG	Clinical remission (CDAI < 150)	62% maintained remission with- out immuno- suppressants	Gastrointestinal toxicity in the conditioning phase
Arruda et al. 2023 [38]	SLE (Brazilian cohort)	Observational	23	6 years	CYC + ATG (low- dose)	SLEDAI = 0; anti-dsDNA negative	57% drug-free remission; nor- malization of complement	No transplant- related deaths; supports safety in developing countries

HSCT hematopoietic stem cell transplantation, *ATG* anti-thymocyte globulin, *CYC* cyclophosphamide, *mRSS* modified Rodnan skin score, *FVC* forced vital capacity, *TRM* transplant-related mortality, *BEAM* BCNU, etoposide, cytarabine, melphalan, *NEDA* no evidence of disease activity, *CDAI* Crohn's Disease Activity Index, *SLEDAI* Systemic Lupus Erythematosus Disease Activity Index

79% in HSCT versus 51% in controls [13]. In addition to the survival benefit, clinically meaningful improvement in pulmonary function and reduction in skin fibrosis were observed, which suggests that the treatment may have a long-lasting impact on the natural history of the disease. Although the primary endpoint of SCOT was overall survival rather than formal remission, the combination of improved long-term survival, stabilization of organ involvement, and reduced need for immunosuppression is consistent with our definition of sustained remission/functional cure in a subset of patients.

Similarly, SLE multicenter studies and international cohorts have shown clinically sustained remission rates of 60% following HSCT with some patients able to stop therapy for more than 10 years [14, 15, 37]. These effects were associated with the disappearance of anti-DNA autoantibodies and normal complement levels. The latter mechanism is suggested to include not only autoreactive B and T clone exhaustion, but also the promotion of regulatory T cells (Tregs) along with immune system functional reprogramming in the tolerogenic mode [38, 39].

However, the operation is not risk-free. The main causes of morbidity are opportunistic infections and toxicities associated with conditioning, which continues to restrict access to HSCT from referral centers in other areas [40]. Nevertheless, reduced mortality and modified conditioning schemes in recent years, such as optimized dosing of cyclophosphamide and antithymocyte globulin, have markedly increased safety and feasibility of the approach [41].

Outside of systemic sclerosis and lupus, HSCT has been studied in multiple sclerosis, idiopathic inflammatory myopathies, as well as Crohn's disease with similar results for patient selections [39, 42]. For example, in multiple sclerosis, HSCT has been demonstrated to result in relapse-free and inflammation-free remission for a period of more than 5 years, even superiority over high-efficacy biological therapies under certain conditions [43].

These findings are in accordance with the idea that HSCT functions as a real “immune reset” resulting in clonal elimination and regeneration of a tolerogenic immune system. Therefore, the

method yields an exceptionally reliable experimental and clinical model of autoimmune reversal—perhaps, as close to a functional cure that can be reported in systemic autoimmune conditions thus far.

Advanced Cellular Therapies: CAR-T Cells and Extracorporeal Photopheresis

The advent of advanced cellular therapies over the last few years has revolutionized thoughts on autoimmune reversal. A type of cellular therapy known as chimeric antigen receptor T-cell therapy, or CAR-T therapy, created primarily for hematologic malignancies, has an outstanding record of performance in severe and aggressive autoimmune diseases that develop resistance to standard-of-care treatments (Table 3).

Chimeric antigen receptor (CAR)-modified T cells that have been genetically engineered to recognize CD19 and other B-cell surface antigens have emerged as a powerful tool for the treatment of refractory B-cell cancers [16]. The pilot study by Mougiakakos et al. consisted of five patients with treatment-resistant SLE who received anti-CD19 CAR-T cells [16]. All five patients achieved complete clinical remission without the need for medications; complement levels and anti-dsDNA antibodies normalized without further recurrence for over 12 months [16]. This case-study marked the beginning of a new age in B-cell-dependent autoimmunity management [16].

In 2024, Schett et al. published a larger series in *Nature Reviews Rheumatology* that included patients with both SLE, Sjögren's syndrome, and autoimmune myopathies, demonstrating the protection and efficacy of anti-CD19 CAR-T treatment in remission induction [36]. The mechanism at play is the selective depletion of autoreactive memory B cells with complementation of a tolerogenic immune system. The regenerative immunity that follows the treatment is accompanied by an increase in regulatory T cells and a decrease in Th17 and Th1 subpopulations, reinstating the damaged homeostasis. New targets are being placed under examination [33, 34].

Table 3 Advanced cellular therapies (CAR-T and extracorporeal photopheresis) in autoimmune diseases

Author/Year [Ref.]	Therapy/Target	Autoimmune disease	Study type	No. of patients	Follow-up	Remission/Outcome definition	Main results	Comments/adverse events
Mougiakakos et al. 2022 [16]	CAR-T cells (anti-CD19)	Systemic lupus erythematosus (SLE, refractory)	Pilot trial	5	12–18 months	Drug-free clinical and serologic remission	100% complete remission; normalization of complement and anti-dsDNA	Mild cytokine release (grade 1–2); no relapse observed
Schert et al. 2024 [17]	CAR-T (anti-CD19)	SLE, Sjögren's syndrome, idiopathic inflammatory myopathies	Multicenter case series	15	12–24 months	EULAR clinical remission criteria	87% sustained remission; restoration of B-cell-naïve phenotype	Evidence of immune tolerance reconstitution
Cao et al. 2023 [16]	CAR-T (anti-BCMA)	Myasthenia gravis	Case report	1	12 months	MG-ADL normalization	Complete clinical remission sustained for 1 year	Reduction of pathogenic plasma cells confirmed
Wang et al. 2023 [42]	CAR-T (anti-CD20)	Neuromyelitis optica spectrum disorder	Pilot study	4	12 months	Absence of relapse; MRI stability	3/4 sustained remission; decreased anti-AQP4 antibodies	Selective depletion of B-cell clones; no severe toxicity
Knobler et al. 2012 [43]	Extracorporeal photopheresis (ECP)	Systemic sclerosis (diffuse)	Randomized controlled trial	64	12 months	mRSS and skin improvement ≥ 25%	60% achieved partial clinical response; safe long-term profile	Demonstrated tolerogenic shift in immune phenotype

Table 3 continued

Author/Year [Ref.]	Therapy/Target	Autoimmune disease	Study type	No. of patients	Follow-up	Remission/Out- come definition	Main results	Comments/ adverse events
Rook et al. 2015 [48]	ECP (UVA + 8-MOP)	Chronic graft-ver- sus-host disease (autoimmunity- like model)	Multicenter cohort	98	24 months	Partial/complete response by NIH criteria	74% overall response rate; steroid-sparing effect	Increased IL-10 and TGF- β ; expansion of Tregs
Scarpato et al. 2020 [49]	ECP	Systemic sclero- sis/autoimmune overlap	Retrospective cohort	21	24 months	Stabilization of pulmonary and cutaneous scores	52% achieved disease stabili- zation; improved microvascular parameters	No severe adverse effects; supports immune toler- ance model

CAR-T chimeric antigen receptor T-cell therapy, *SLE* systemic lupus erythematosus, *ECP* extracorporeal photopheresis, *BCMA* B-cell maturation antigen, *MG-ADL* Myasthenia Gravis Activities of Daily Living scale, *AQP4* aquaporin-4, *mRSS* modified Rodnan skin score, *8-MOP* 8-methoxypsoralen, *NIH* National Institutes of Health, *Tregs* regulatory T cells, *IL-10* interleukin-10, *TGF- β* transforming growth factor beta

Early-stage clinical trials testing anti-BCMA CAR-T cells in myasthenia gravis, neuromyelitis optica, and rheumatoid arthritis have reported sustained remission, characterized by stabilization and durable normalization of autoantibody levels [16, 17, 36, 49]. This approach is considered a highly selective immunologic reset, aiming to eradicate autoreactive B-cell clones while preserving immune competence [36, 49, 68].

In parallel, extracorporeal photopheresis (ECP) has been investigated as immunomodulatory therapy. During ECP, leukocytes collected via leukapheresis are treated with 8-methoxypsoralen and UVA light before reinfusion [20, 37]. The resulting immunologic effects include the induction of tolerogenic dendritic cells and the expansion of CD4⁺CD25⁺FoxP3⁺ regulatory T cells, accompanied by increased IL-10 and TGF- β production [59, 62, 68]. Clinical studies in systemic sclerosis have shown reductions in skin thickness and improvements in microvascular function [20, 37]. Open-label cohorts suggest that ECP may promote long-term clinical stability in refractory autoimmune diseases [62, 68]. Both CAR-T and ECP ultimately converge on the same therapeutic objective: the restoration of durable immune tolerance [36, 49, 68].

Nonetheless, these therapies differ in their mechanisms of inducing tolerance. CAR-T cells operate through targeted depletion of autoreactive B-cell populations and modulation of T–B interaction pathways, leading to profound immunologic reconstitution [16, 17, 36, 49]. In contrast, extracorporeal photopheresis promotes peripheral tolerance more gradually, by inducing apoptosis of pathogenic lymphocytes and enhancing regulatory immune pathways [20, 37, 62]. Despite these differences, median long-term outcomes—characterized by sustained clinical remission, serologic normalization, and prolonged freedom from autoreactive B cells—support the concept that cellular therapies may induce a functional cure in a select subset of patients with severe, treatment-refractory autoimmune diseases [36, 49, 68].

Environmental and Nutritional Interventions

Environmental and nutritional factors play a central role in modulating autoimmunity. Among them, vitamin D stands out for its broad immunoregulatory effects. It acts through receptors expressed on dendritic cells and lymphocytes T and B, promoting differentiation of regulatory T cells (Tregs) and suppressing pro-inflammatory responses mediated by IL-17 and IFN- γ [18, 44, 51].

Longitudinal studies and meta-analyses show that serum 25-hydroxyvitamin D levels above 40 ng/ml are associated with lower disease activity in SLE, multiple sclerosis (MS), and rheumatoid arthritis (RA) [18, 52]. Furthermore, controlled clinical trials have demonstrated that supplementation can reduce relapse frequency and enable tapering of corticosteroids or immunosuppressants [53]. Although not all patients achieve complete remission, some reports document sustained, drug-free remissions—a form of biochemical functional cure mediated by immune re-education.

Another growing area of investigation involves dietary patterns and the intestinal microbiota. Whole-food, plant-based diets low in saturated fats and rich in fermentable fibers have been shown to modulate gut microbiota, promoting greater bacterial diversity and increased abundance of short-chain fatty acid (SCFA)-producing species such as *Faecalibacterium prausnitzii* and *Roseburia* spp. [19, 54]. These metabolites—particularly butyrate and propionate—induce Treg differentiation and suppress dendritic cell and Th17 activation, promoting peripheral tolerance [55].

In parallel, recent studies of fecal microbiota transplantation (FMT) in ulcerative colitis, RA, and SS suggest that intestinal recolonization with SCFA-producing strains can reduce inflammatory activity and induce prolonged remissions [45, 56]. Although the evidence remains preliminary and based on small series, these findings support the view that the microbiota functions as an adaptive immunologic organ, whose reprogramming may represent a natural form of immune reset [57] (Table 4).

Beyond nutrition, environmental and behavioral factors—such as sunlight exposure, regular physical activity, and circadian rhythm regulation—deeply influence immune balance. Sunlight not only enhances vitamin D synthesis but also regulates melatonin and cortisol production, hormones that modulate inflammatory reactivity [46, 58]. Observational studies indicate that regular exercise reduces IL-6 and TNF- α levels, improves immune plasticity, and is associated with longer periods of clinical remission [59].

These observations support the concept that the metabolic and immunologic environment is a key determinant of autoimmunity reversibility. Nutritional and lifestyle interventions, even as adjuvants, can act as catalysts of functional cure, potentiating the effects of more specific immunologic therapies.

Paraneoplastic Syndromes and Remission After Cancer Treatment

Paraneoplastic syndromes represent a striking example of how resolution of a tumor-associated antigenic stimulus can lead to complete remission of autoimmunity. In these cases, the immune response directed against tumor antigens cross-reacts with self-proteins, breaking tolerance and leading to autoimmune manifestations. Thus, elimination of the tumor—whether through surgical resection, chemotherapy, or radiotherapy—can restore immune homeostasis and halt autoimmune aggression [10, 11, 16, 47].

Dermatomyositis associated with solid tumors is the most well-documented paradigm of this relationship. Studies indicate that up to 30% of dermatomyositis cases are associated with occult malignancies, mainly of the ovary, lung, and gastrointestinal tract [29, 60]. Aberrant expression of antigens such as TIF1 γ , NXP2, and Mi-2 in tumor cells triggers a cross-reactive immune response against muscle and skin. Recent systematic reviews confirm that effective treatment of the primary tumor often results in complete dermatomyositis remission, particularly when oncologic therapy is initiated early [60].

Table 4 Environmental and nutritional interventions promoting immune tolerance and sustained remission in autoimmune diseases

Author/Year [Ref.]	Intervention/ exposure	Autoimmune disease	Study type	N (patients)	Follow-up	Outcome defini- tion	Main results	Comments/mech- anistic insights
Aranow et al. 2011 [18]	Vitamin D sup- plementation (4000 IU/day)	Systemic lupus erythematosus (SLE)	Randomized controlled trial	267	12 months	Reduction in SLEDAI score; ↑Tregs	Significant disease activity reduc- tion; improved Treg/Th17 ratio	Dose-dependent immunomodula- tory effect
Smolders et al. 2017 [52]	High-dose vitamin D3 (10,000 IU/ day)	Multiple sclerosis	Prospective cohort	145	24 months	Relapse rate; MRI lesions	Fewer relapses and new MRI lesions vs. con- trols	Supports vitamin D neuro-immu- noregulation
de Souza et al. 2020 [53]	Vitamin D optimization (≥ 40 ng/ml)	Rheumatoid arthritis	Interventional open-label	62	18 months	DAS28 < 2.6 (remission)	35% achieved sustained remis- sion; steroid tapering pos- sible	Positive correla- tion between serum 25(OH) D and remis- sion
McDougall et al. 2018 [19]	Whole-food, plant-based diet	SLE/Sjögren's syndrome	Prospective pilot study	36	12 months	Clinical remis- sion/symptom- free status	28% complete remission; 44% symptom reduc- tion	Increased SCFA- producing gut microbiota
Chiang et al. 2021 [54]	High-fiber prebi- otic diet	Rheumatoid arthritis	Randomized controlled trial	60	6 months	CRP and DAS28 reduction	↓IL-6 and TNF- α; ↑butyrate and propionate levels	Demonstrated metabolic- immune cross- talk
Zhang et al. 2023 [56]	Fecal microbiota transplantation (FMT)	Ulcerative colitis	Phase II clinical trial	73	12 months	Clinical remission (Mayo ≤ 2)	32% sustained remission vs. 9% placebo	Restoration of SCFA-produc- ing bacteria

Table 4 continued

Author/Year [Ref.]	Intervention/ exposure	Autoimmune disease	Study type	N (patients)	Follow-up	Outcome defini- tion	Main results	Comments/mech- anistic insights
Wang et al. 2024 [45]	FMT (autologous donors)	Rheumatoid arthritis	Case series	10	12 months	DAS28 reduc- tion ≥ 1.2 ; drug- free status	4/10 achieved drug-free remission; $\downarrow$ Th17/ $\uparrow$ Treg balance	Supports micro- biota-immune reprogramming hypothesis
Gomes et al. 2022 [59]	Aerobic physical exercise	Autoimmune rheumatic diseases	Systematic review/meta- analysis	1250	Variable	$\downarrow$ IL-6, TNF- α , improved disease activity indices	Improved immune flexibil- ity and longer remission	Confirms anti- inflammatory exercise effects

SLEDAI Systemic Lupus Erythematosus Disease Activity Index, *Tregs* regulatory T cells, *Th17* T-helper 17 cells, *MRI* magnetic resonance imaging, *DAS28* Disease Activity Score in 28 joints, *SCFA* short-chain fatty acids, *CRP* C-reactive protein, *TNF- α* tumor necrosis factor alpha, *IL-6* interleukin-6, *FMT* fecal microbiota transplantation

Table 5 Paraneoplastic syndromes: remission of autoimmunity following cancer treatment

Author/Year [Ref.]	Neoplasia/Tumor condition	Paraneoplastic autoimmunity	Study type	N	Oncologic treat- ment (main)	Autoimmune outcome (crite- rion)	Key results	Remarks/adverse events
Selva- O'Callaghan et al. 2022 [10]	Various solid tumors (ovary, lung, GI, breast, etc.)	Dermatomyositis/ cancer-associ- ated myositis	Narrative review/ therapeutic overview	—	Surgical resection, chemotherapy, and/or radio- therapy	Clinical remis- sion of myosi- tis (muscle strength, CK) and skin disease	Effective tumor control fre- quently paral- leled myositis remission; mul- tidisciplinary onco-rheuma- tology approach improved outcomes	Aggregated evi- dence; heteroge- neous remission definitions
Fujino et al. 2024 [11]	Breast carcinoma	Dermatomyositis	Case report	1	Curative surgery (mastectomy)	Complete and sustained clini- cal remission	Notable remission of dermato- myositis after tumor removal, with full clinical and laboratory normalization	No significant oncologic AEs reported
Teboul et al. 2025 [41]	Solid neoplasms (paraneoplastic cohort)	Cancer-associated dermatomyositis	Multicenter retro- spective cohort	73	Tumor-specific therapy accord- ing to histology/ stage	Myositis response (muscle strength, enzymes, skin)	Prognostic factors for remission identified; autoimmune response corre- lated with tumor control and early treatment	Variable regi- mens; highlights need for system- atic malignancy screening

Table 5 continued

Author/Year [Ref.]	Neoplasia/Tumor condition	Paraneoplastic autoimmunity	Study type	N	Oncologic treat- ment (main)	Autoimmune outcome (crite- rion)	Key results	Remarks /adverse events
Raja et al. 2022 [42]	Thymoma (subset in MG database)	Thymoma-associ- ated myasthenia gravis	Database analysis (perioperative outcomes)	—	Thymectomy	Sustained clinical improvement/ remission	Evidence supports thymectomy safety and long-term MG control; indirect evidence for durable remis- sion	Study focused on periopera- tive outcomes; not designed for long-term remission rates
Snowden et al. 1997 [30]	Non-Hodgkin lymphoma	Systemic lupus erythematosus (coexistent)	Case report	1	Chemotherapy for lym- phoma + autolo- gous BMT	Complete and prolonged remission of SLE	Long-term lupus remission after antineoplastic therapy; illus- trates elimina- tion of tumor- driven immune stimulation	Intensive regimen; onco-hematologic context

CK creatine kinase, *DM* dermatomyositis, *GI* gastrointestinal, *SLE* systemic lupus erythematosus, *MG* myasthenia gravis, *BMT* bone marrow transplantation

Similarly, thymoma-associated myasthenia gravis constitutes another classic model of neoplasia-induced autoimmunity reversal. The thymoma expresses muscle autoantigens that stimulate the production of autoantibodies against the acetylcholine receptor. Thymectomy—especially when performed early—can lead to sustained clinical remission and, in some cases, to a functional cure characterized by absence of relapse and complete withdrawal of immunosuppressants [47, 61].

There are also reports of autoimmune hemolytic anemia and immune thrombocytopenic purpura (ITP) secondary to lymphomas achieving complete remission after chemotherapy or allogeneic bone marrow transplantation, suggesting that elimination of the neoplastic clone removes the source of chronic stimulation of autoreactive B cells [16, 62]. The same phenomenon has been described in patients with concurrent SLE and lymphoma, in whom chemotherapy led to simultaneous regression of both diseases [32] (Table 5).

These observations reinforce the biological principle that autoimmunity can be reversible when its immunogenic trigger is eliminated. The disappearance of the autoreactive response following cancer cure provides an experimental model of immune reprogramming analogous to that observed after eradication of chronic infections such as *Helicobacter pylori* or hepatotropic viruses [25, 34].

Mechanistically, tumor resolution is accompanied by reductions in type I interferons, normalization of PD-1 and CTLA-4 expression, and reversal of the Th1/Th17 inflammatory profile toward a tolerance state mediated by Tregs [63, 64]. These same mechanisms explain why modern oncologic immunotherapies, such as immune-checkpoint inhibitors, can induce de novo autoimmune diseases—the so-called immune-related adverse events (irAEs)—illustrating how narrow the boundary is between antitumor immunity and autoimmunity [47, 65].

Interestingly, the reverse also applies: immunomodulatory therapies used in autoimmunity—such as HSCT and CAR-T cells—can eliminate residual neoplastic clones, highlighting a bidirectional axis between autoimmunity and oncogenesis [17, 42, 63]. This dual role of the

immune system as both a tumor-eliminating and autoimmunity-inducing agent exemplifies the delicate equilibrium between immune surveillance and tolerance that is essential for immunologic homeostasis.

DISCUSSION

The possibility of a functional cure in autoimmune diseases has ceased to be merely theoretical as different interventions have demonstrated the capacity to re-establish immunological tolerance and interrupt the cycle of self-aggression. Historically, autoimmunity was considered irreversible because it involves disruption of central and peripheral tolerance, with expansion of autoreactive T- and B-cell clones and cumulative tissue damage [1, 2]. However, advances in immunology, cell biology, and targeted therapies have shown that, under certain conditions, the immune system can be “rebooted” and return to a functional equilibrium similar to that observed in healthy individuals [3, 4].

The first line of evidence comes from the eradication of infectious triggers, especially *Helicobacter pylori*. The most established paradigm is immune thrombocytopenic purpura (ITP), in which the microorganism appears to act as a cofactor in triggering and perpetuating the autoreactive response against platelets. Meta-analyses show response rates exceeding 50% after eradication, often with sustained normalization of platelet counts and discontinuation of immunosuppressants [8, 9, 25]. The response is particularly marked in Asian and European countries, suggesting that genetic differences in the recognition of bacterial antigens and Fcγ receptor polymorphisms may modulate outcomes [8].

The most accepted pathophysiological mechanism is molecular mimicry, whereby *H. pylori* epitopes share structural homology with platelet glycoproteins (such as GPIIb/IIIa), leading to the production of autoantibodies. Bacterial elimination halts the antigenic stimulus and allows regulatory mechanisms—particularly regulatory T cells (Tregs) and IL-10 production—to restore peripheral tolerance [26–28]. This phenomenon

also applies to other diseases, such as psoriasis and lichen planus, in which the pathogen acts as a perpetuator of chronic inflammation [26–28].

In Behçet's disease, clinical improvement after *H. pylori* eradication reinforces the hypothesis that infectious agents maintain an abnormal immune response through continuous activation of Th1 and Th17 T cells [29]. Reports of remission in autoimmune thyroiditis after antibacterial therapy, although rare, suggest that the microbiota–mucosa axis may be crucial for systemic immune balance [31]. Although fragmented, these findings point to a fundamental principle: autoimmunity can be reversible when the antigenic trigger is removed early. This model underscores the role of chronic exposure to environmental antigens in maintaining the autoreactive state, offering a conceptual analogy to remission observed after viral eradication (such as chronic hepatitis) in other autoimmune conditions [6, 62].

The second line of evidence concerns immune reset achieved through chemotherapy regimens and autologous hematopoietic stem cell transplantation (HSCT). The ASTIS and SCOT trials in systemic sclerosis were landmarks in this field, showing that HSCT not only controls inflammation but also modifies the natural history of the disease [12, 13]. Ablative immunosuppression eliminates autoreactive clones, and subsequent regeneration of the lymphocyte repertoire from hematopoietic progenitors rebuilds a tolerogenic immune network [35, 51]. Immunophenotypic studies confirm that, after HSCT, there is an increase in Tregs, a decrease in Th17 cells, and normalization of the memory B-cell repertoire, which correlates with sustained clinical remission [58–60].

In SLE, this phenomenon is particularly notable. International cohorts show that up to 60% of patients achieve long-term drug-free remission, some for more than 10 years, with the disappearance of anti-DNA autoantibodies and normalization of complement [14–16]. These results support the hypothesis that, once “restarted,” the immune system can restore mechanisms of self-tolerance. Even so, early mortality related to conditioning regimen toxicity and opportunistic infections remains the main barrier to wider adoption [51].

In diseases such as multiple sclerosis, inflammatory myopathies, and Crohn's disease, the same principles apply. Immune reset allows reconfiguration of the balance between effector and regulatory cells, reducing autoreactivity and promoting immune re-education [33–35]. Experimental models of autoimmunity corroborate these findings, suggesting that after selective destruction of pathogenic T-cell clones and thymic regeneration, central tolerance is restored [57].

Advanced cellular therapies represent one of the most significant advances in the quest to restore immune tolerance. The introduction of CAR-T cells) targeting CD19—originally developed for hematologic malignancies—has opened a new perspective in the treatment of severe, refractory autoimmune diseases [16, 17]. The pioneering study by Mougiakakos et al. demonstrated that selective depletion of CD19⁺ B cells by CAR-T can abolish autoantibody production and induce sustained remission in patients with refractory SLE [16]. In 2024, Schett and colleagues expanded this experience to a larger series involving SLE, autoimmune myopathies, and SS, confirming the efficacy and safety of the approach [17]. The pathophysiological explanation involves eradication of autoreactive memory B-cell clones and regeneration of a tolerogenic B-cell repertoire [48].

This reconfiguration of the B-cell compartment is accompanied by restoration of Treg function and reduction of Th17 populations—both central to autoimmune inflammation [58, 60]. The phenomenon is analogous, albeit more selective, to what is observed after HSCT, constituting an immune “reset” without global ablative conditioning [48, 61]. In addition, anti-CD19 CAR-T cells appear to indirectly modulate costimulatory axes such as CD40/CD40L, interfering with T–B interaction and durably reducing autoimmunity [47]. Expansion of this technology includes the development of anti-BCMA and anti-CD20 CAR-T products, in early phases for diseases such as myasthenia gravis and neuromyelitis optica; although still preliminary, there are reports of sustained remission with follow-up of up to 2 years [48].

Another cellular approach, older and with a distinct immunomodulatory profile, is

extracorporeal photopheresis (ECP). This procedure—which combines leukocyte collection, exposure to 8-methoxypsoralen and UVA light, and subsequent reinfusion—induces apoptosis of pathogenic lymphocytes and promotes immune tolerance [20, 37]. Trials and series in systemic sclerosis have shown clinical stabilization and microvascular improvement, albeit without robust superiority across all endpoints [20, 37]. Mechanistically, there is evidence of expansion of CD4⁺CD25⁺FoxP3⁺ Tregs and increased IL-10/TGF- β , cytokines associated with peripheral tolerance [58]. Although ECP rarely leads to complete remission, it represents a physiological model of gradual tolerance induction that may be replicated by new cellular and immunomodulatory therapies [61].

In the environmental and nutritional axis, vitamin D deserves emphasis for its multifaceted immunoregulatory role. In addition to promoting Treg differentiation and reducing expression of pro-inflammatory cytokines such as IL-17 and IFN- γ , it acts on the transcription of genes associated with tolerance [18, 38]. Clinical evidence suggests reduced activity in multiple autoimmune diseases and, in some cases, sustained remissions with prolonged supplementation [18]. Beyond vitamin D, dietary factors and the intestinal microbiota also appear to influence remission. Plant-based diets with lower saturated fat content modulate the microbiota and the production of short-chain fatty acids (SCFAs), exerting tolerogenic effects on Tregs and dendritic cells [19, 50, 56]. Studies of fecal microbiota transplantation (FMT) in ulcerative colitis and RA show promising results, with reports of sustained remission in subgroups [39]. Although still incipient, these data support the concept that the microbiota functions as an adaptive immunologic organ whose reprogramming may be equivalent to an immune “reset” [50, 56].

Finally, behavioral factors such as regular physical activity also modulate immunity; meta-analyses in autoimmune rheumatic diseases indicate reductions in systemic inflammation and potentially increased immunologic plasticity, favoring prolonged periods of remission [40].

Paraneoplastic syndromes constitute one of the most eloquent examples of the interconnection between antitumor immune responses and

autoimmunity. In these settings, autoreactive activity arises as a consequence of a vigorous immune response against tumor antigens that mimic self-proteins. Thus, elimination of the tumor—via surgical resection, chemotherapy, or transplantation—often leads to resolution of the associated autoimmune manifestations [10, 11]. Dermatomyositis associated with solid tumors is the most studied model. Aberrant expression of antigens such as TIF1 γ and NXP2 in tumors triggers cross-reactive immune responses; studies and clinical series show dermatomyositis remission after oncologic treatment, especially when the tumor is addressed early [10, 11, 41]. Thymoma-associated myasthenia gravis shares similar mechanisms, and thymectomy can result in sustained clinical remission [42].

These observations suggest that, in certain contexts, autoimmunity is an epiphenomenon of an antitumor response directed against shared antigens. When the tumoral stimulus disappears, the autoreactive response ceases—a principle analogous to that observed with eradication of chronic infections such as *H. pylori* [6, 25]. Cases of autoimmune hemolytic anemia and thrombocytopenic purpura secondary to lymphoma that enter remission after chemotherapy or allogeneic transplantation reinforce the notion that elimination of the neoplastic clone reduces chronic stimulation of autoreactive B cells [32]. Strikingly, there are reports of patients with concomitant SLE and lymphoma who achieved remission of both conditions after chemotherapy [30]. These findings reaffirm that the mechanisms shared between antitumor surveillance and loss of tolerance involve type I interferon pathways, immune checkpoints (CTLA-4/PD-1), and remodeling of the immune microenvironment [43, 44].

Study of these phenomena provides valuable insights into the balance between protective immunity and autoimmunity. Oncologic immunotherapies, such as checkpoint inhibitors, show that hyper-stimulation of antitumor immunity can induce de novo autoimmune diseases (immune-related adverse events, irAEs), highlighting how tenuous the boundaries are between cure and self-aggression [45, 52]. Interestingly, the reverse is also true: interventions that restore tolerance (such as

anti-CD19 CAR-T cells or, in certain contexts, HSCT) may, in some cases, eliminate residual tumor clones—a bidirectional axis between autoimmunity and oncogenesis [17, 51].

This review has methodological limitations. First, most available evidence derives from case series and retrospective cohorts, which restricts generalizability. In addition, the definition of functional cure is not yet universally accepted, varying by disease and by the remission parameters used [22, 23]. Moreover, remission outcomes reported in the original trials and cohorts were heterogeneous; wherever possible we reclassified them according to current consensus definitions (DAS28, DORIS, EUSTAR), but this introduces some unavoidable imprecision when comparing across studies and therapeutic axes [66]. Heterogeneous follow-up among studies and the absence of biomarker standardization also hinder comparisons across different therapeutic axes [67].

Nevertheless, the strengths of this work lie in its breadth and conceptual integration: distinct approaches—infection-related, immunologic, cellular, and environmental—were analyzed within a single theoretical framework, that of autoimmunity reversibility [68]. This cross-cutting perspective allows us to understand that, regardless of the therapeutic method, functional cure depends on restoration of immunological tolerance, whether through elimination of antigenic triggers, reconfiguration of the lymphocyte repertoire, or environmental and metabolic modulation [67].

Prospects for the future of functional cure in autoimmune diseases are promising. Next-generation trials are already exploring combinations of cellular and biological strategies aimed at sustained remission with low toxicity [48]. The use of engineered regulatory T cells (CAR-Tregs) shows potential to induce antigen-specific tolerance in diseases such as type 1 diabetes and multiple sclerosis [58]. In parallel, epigenetic and metabolomic approaches are beginning to elucidate how environmental and nutritional factors shape the immune response and can be manipulated therapeutically [49]. Reprogramming of the intestinal microbiota, combined with immunologic interventions, is emerging as an integrative pillar between

immunotherapy and preventive medicine [50, 56].

In the long term, redefining functional cure in autoimmune diseases will require objective, consensus parameters integrating clinical, serologic, immunologic, and molecular aspects. Thus, cure will cease to be a binary concept and will come to represent a continuum of functional restoration of the immune system.

CONCLUSIONS

The findings of this review show that functional cure in autoimmune diseases, though rare, is a biologically plausible phenomenon documented across multiple clinical contexts. Evidence indicates that autoimmunity can be reversed when its sustaining mechanisms—infection-related, cellular, metabolic, or neoplastic—are interrupted. Future studies are necessary with standardized definitions, predictive biomarkers, and long-term trials to expand the clinical application of this emerging paradigm.

ACKNOWLEDGEMENTS

Medical writing assistance. AI-assisted software (ChatGPT, OpenAI) was used exclusively for language editing and to improve clarity. No content generation, data extraction, or reference creation was performed using AI tools. The authors take full responsibility for the manuscript.

Author Contributions. Jozélio Freire de Carvalho and Carlos Ewerton Maia Rodrigues contributed to the conception and design of the study, literature search, data extraction and interpretation, and drafting and critical revision of the manuscript. Both authors approved the final version and agree to be accountable for all aspects of the work.

Funding. No funding or sponsorship was received for this study or for the publication of this article.

Data Availability. All data generated or analyzed during this study are derived from previously published articles that are cited in the reference list and are available in the public domain.

Declarations

Conflict of Interest. Carlos Ewerton Maia Rodrigues has nothing to disclose. Jozélio Freire de Carvalho is a member of the Editorial Board of *Rheumatology and Therapy* and had no involvement in editorial decision-making regarding this manuscript.

Ethical approval. Not applicable. This study is based exclusively on previously published literature and does not involve human participants, animals, or identifiable data.

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